News

A lonely USGS scientist stares out over miles of whiteness in the antarctic. Observations are impaired by blowing snow which results in poorly defined horizons. (USGS photo)

U.S. Team Mapping Last of Antarctica's Unexplored Regions. One of the most remote coastal regions in Antarctica is now being mapped and its geology described by a team of U.S. Geological Survey engineers and scientists.

The first season's exploration of a planned 3-year project was conducted during the 3-month period from November 1969 to February 1970 along the Lassiter Coast of the Antarctic Peninsula—that narrow finger of land jutting northward toward South America. The area explored included the Latady, Scaife, Wilkins, and Hauberg mountains, the Rare range, and parts of Wetmore glacier, an area that totals more than 6000 square miles. Aerial photographs taken of this region in 1967 were used to compile a sketch map that served as a guide to the explorers.

Paul L. Williams, a geologist with the USGS and leader of the 8-man team, said that when aircraft landed his party at the north end of the Wetmore glacier on November 9, 1969, it marked the first time that man had set foot in the area. 'The nearest approach to this area was made when a combined British-American sledge party traversed the sea ice to Bowman Peninsula, about 40 miles away,' he said.

During the total of 91 working days the party was often handicapped by prolonged periods of 'white out' and blowing snow. The temperature in the area ranged from a low of -30° F to a high of +30° F.

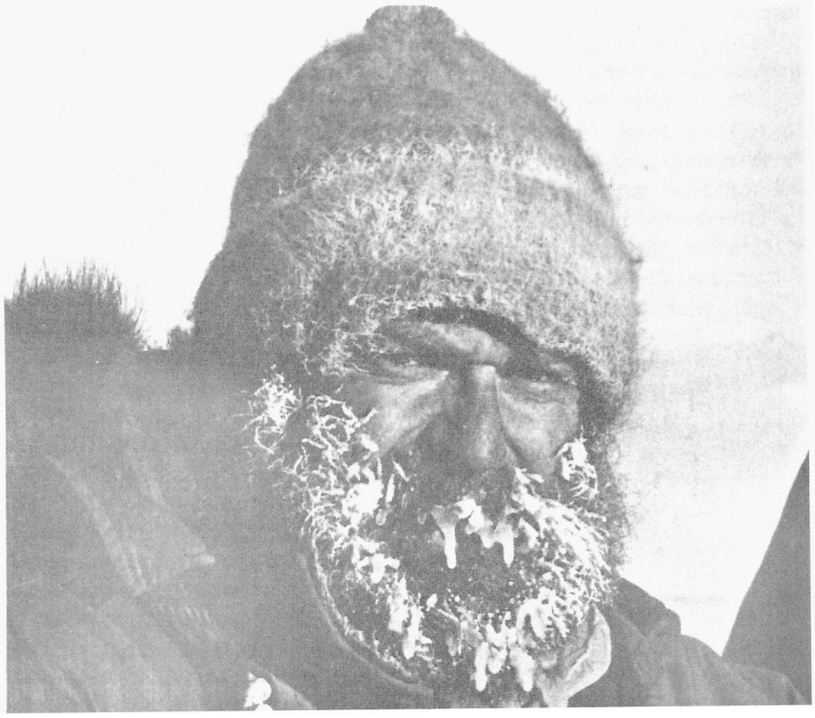
Eberhard G. Schirmacher, who served as the topographic engineer-in-charge said that 'to establish the needed geodetic control network throughout the 6000 square miles region the engineers first established an initial position by daylight star observation.' He said that they were later able to complete about 140 miles of electronic traverse southward from the initial point, establish 12 geodetic control stations, and set 9 permanent survey markers in rock outcrop. The information will be used in the future during the construction of a series of topographic reconnaissance maps of the area at a scale of 1:250,000.

The topographic engineers also erected a series of posts in the ice of 3/4-mile intervals across the 7-mile profile of the Wetmore glacier near their base camp. Called a 'strain net,' the horizontal line of posts acts as ice-movement markers. By noting changes in the positions of these markers in subsequent surveys, scientists will obtain information about the activity of that part of the great antarctic ice sheet.

Williams will lead his second expedition to the Lassiter coast this fall, with Schirmacher again his deputy and engineer-in-charge.

Pollution Study in the Antarctic. Six United States scientists have traveled to the antarctic to conduct research related to the global distribution of

The frost-bearded Paul L. Williams, leader of the 8-man USGS team in the antarctic. (Navy photo)



pollutants, including lead and DDT.

The scientists, from the University of Rhode Island and the University of California, Berkeley, will participate in the U.S. Antarctic Research Program (USARP), of the National Science Foundation.

Traces of lead in samples of antarctic ice were found several years ago, as were traces of DDT in penguins. This year three scientists from the University of Rhode Island will sample the relatively pure antarctic atmosphere for traces of lead, copper, iron, aluminum, and other metals and compare the concentrations with those found in the North and South Atlantic Oceans and in Hawaii. The comparisons are expected to provide valuable information on the possible global transport of such pollutants.

Reproductive failure due to eggshell thinning caused by the accumulation of chlorinated hydrocarbons such as DDT has almost exterminated the Brown Pelican of Louisiana and southern California. Such eggshell thinning has also been found in a number of other species during the past year. Three University of California, Berkeley, scientists hope to gather additional information about the distribution patterns of DDT and other chlorinated hydrocarbons in several areas of the oceans, studying the reproductive success of a wide-ranging petrel known to be accumulating large quantities of chlorinated hydrocarbons in its system. The scientists believe the bird, the Wilson's Petrel, may be used to monitor such pollution because it is so numerous and because it ranges from north to south polar regions. The scientists will also study the distribution of polychlorinated biphenyls, used in industry as plasticizers and as additives to some paints, lubricants, and insulators.

The two projects are part of the USARP program, which this year includes nearly 200 scientists and technicians from throughout the United States who will fan out across Antarctica during the next several months to learn more about the frozen seventh continent.

Coming from more than 50 colleges and universities, Government agencies, and industrial firms, they are participating in the National Science Foundation-supported Antarctic Research Program for 1970-1971.

For the second year the program will include women researchers. One, a physicist from the University of Washington, Seattle, will conduct research on the earth's ionosphere from a small scientific outpost near Byrd station. The other, a biologist from Utah State University, will return to Antarctica for the second season to assist in a study of anti-predation behavior of the Adelie penguin.

Approximately $7.5 million in NSF grants and contracts will support the scientific investigations to be conducted on the continent, from research vessels, and in laboratories in the United States.

Continuing a highly successful project begun last season, 12 investigators from Ohio State University will conduct studies in geology and geophysics and will search for additional fossils of land vertebrates in the Central Transantarctic Mountains. During the last antarctic summer (winter in the Northern Hemisphere) the group, with other investigators, found large numbers of such fossils giving great support to the long-controversial theory of continental drift. The find was labeled one of the outstanding paleontological discoveries of the century.

In what may be one of the last major reconnaissance field studies of the U.S. research program, four geologists and four topographic engineers of the U.S. Geological Survey, Department of the Interior, will use motor toboggans for transport in examining the geology of the Lassiter Coast area on the east side of the Antarctic Peninsula and to establish ground control for aerial mapping. Such reconnaissance work is considered essentially completed in Antarctica and investigations in recent years have emphasized more specialized studies.

On the stark white snow plains near Byrd station, University of Denver scientists will launch instrumented balloons to study cosmic radiation impinging upon the earth's high atmosphere. Such radiation, propagated along the lines of force of the earth's magnetic field, is more intense in the polar areas, where the lines of force are almost vertical.

On sea ice near McMurdo station, scientists from the University of Minnesota will use radio transmitters and underwater television in a study of the population dynamics of antarctic seals.

A Stanford University team, operating at Siple station, will construct a 13-mile long dipole antenna in a study of the earth's plasmapause, a region of anomolously low ion density which is the upper boundary of the more dense plasmasphere. Siple station was established last year at the base of the Antarctic Peninsula.

The 1970-1971 United States Antarctic Research Program will be conducted at six United States antarctic stations and at several remote field sites, aboard the research ship *Eltanin* in the Indian Ocean, on the research ship *Hero* operating in Antarctic Peninsula waters, aboard icebreakers in the Ross Sea and, through arrangements with the Soviet Union, at Vostok station. U.S. stations operating throughout the year are: Byrd, McMurdo, Palmer, and South Pole. Siple station, near the base of the Antarctic Peninsula, will be opened during the Antarctic summer from November through January. Hallett station, site of U.S. and New Zealand scientific programs, is open from October through February. Biological work is concentrated at the coastal stations—McMurdo, Hallett, and Palmer. Research in upper atmosphere physics is concentrated at Byrd, Siple, and South Pole stations. The majority of U.S. scientists will conduct research on the continent during the summer months; about 30 will remain during the winter.

U.S.—Japan Cooperative Ice Study. The Ocean-Atmosphere Research Institute, on recommendation of the United States-Japan Committee on Scientific Cooperation has received a grant from the National Science Foundation in partial support of a study of large-scale sea-ice, sea temperature, and weather interrelations, and their significance for long-range foreshadowing.

A team of Japanese oceanographers and meteorologists headed by Kantaro Watanabe, Chief, Oceanographic Section, Kobe Marine Observatory, will cooperate with an American group headed by Irving I. Schell, Sen-